Deep Unfolding Network Enhanced by Transformer Priors for Unregistered Hyperspectral and Multispectral Image Fusion

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***Abstract*— In satellite remote sensing, the complementary nature of hyperspectral (HSI) and multispectral (MSI) imagery necessitates their fusion to enhance both spatial and spectral resolution. However, the inherent misalignment between these datasets, due to differences in acquisition conditions, poses a significant challenge. This study presents a novel approach, the deep unfolding network enhanced by Transformer priors (DUNET), to address the simultaneous registration and fusion of HSI and MSI. Unlike conventional deep fusion methods, which are often treated as opaque "black boxes," DUNET incorporates the deep unfolding method, leveraging mutual information and deep priors to facilitate a better degradation model-informed fusion process. The proposed network incorporates hybrid attention Transformers (HATs) and spatial-frequency modules to fully exploit the spatial–spectral information of HSI, resulting in a more accurate and detailed representation of the scene. We conducted extensive quantitative and visual experiments on three standard HSI datasets. The results demonstrate that our proposed DUNET method outperforms the existing mainstream algorithms in the field of remote sensing image fusion, showcasing its effectiveness. Specifically, our proposed method achieves the improvements of 3.4, 5.1, and 8.2 dB in terms of peak signal-to-noise ratio (PSNR) compared with the latest methods on the ICVL, Chikusei, and Houston datasets, respectively.**

***Index Terms*— Hyperspectral image (HSI) fusion, image prior, nonregistered, Transformer.**

NOMENCLATURE

$\|\|\cdot\|\|_F^2$	Frobenius norm.
$\|\|\cdot\|\|_1$	$L1$ norm.
$\mathbf{X}$, $\mathbf{Y}$, $\mathbf{Z}$	Matrix form of variable.
$\mathcal{X}$, $\mathcal{Y}$, $\mathcal{Z}$	Tensor form of variable.
α, β, γ, δ	Regularization parameter.
$\mathbf{C}$, $\mathbf{R}$	Degradation functions in the spatial and spectral domains.
$\mathcal{X}_n$	nth iteration of $\mathcal{X}$.
$(.)^T$	Transpose.

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I. INTRODUCTION

HYPERSPECTRAL images (HSIs) provide advantages in identifying the physical properties of object materials compared with commonly used multispectral images (MSIs) [1] by capturing a large amount of narrowband spectral information, resulting in richer spectral features. In contrast, MSI typically contains only a small number of wideband spectral information, thus limiting its capability in this aspect. HSI has made significant advances in the field of computer vision, successfully applied to image segmentation [2] and object recognition [3], utilizing its spectral data to accurately distinguish and identify objects in complex scenes. Although HSIs achieve high spectral resolution by utilizing contiguous and narrowbands, their spatial resolution often falls short of that found in everyday RGB images. This discrepancy arises from the dense spectral bands present in hyperspectral sensors, leading to a limited number of photons received on average per band. The low spatial resolution of captured HSIs severely impacts their practical value. The natural solution to overcome this limitation involves simultaneously capturing LR-HSI and HR-MSI, such as RGB images, and merging them to create images with both high spatial and spectral resolution. This procedure is commonly termed HSI and MSI fusion (HMF) or hyperspectral super-resolution [4], [5].

Traditional methods address the ill-posed nature of HMF problems by incorporating prior information obtained from HSI into regularization terms. These priors include techniques, such as sparse representation [6], [7], [8], low-rank assumptions [9], [10], [11], nonlocality [12], and self-similarity [13], [14]. Nevertheless, manually predefined priors are subject to inherent limitations, demanding domain expertise for practical implementation and potentially lacking in the comprehensive capture of all data features. Consequently, this constrains the efficacy and generalization potential of optimization-based methodologies. With recent advancements in deep learning, some researchers have turned to using deep learning to address fusion-based HMF problems, significantly enhancing fusion performance by learning data-driven priors from extensive datasets. Researchers have proposed specialized modules, such as spatial, spectral, and spatiospectral attention [15], to fully utilize the features of HSI data [16]. Meanwhile, a series of HMF methods based on unfolding networks [17], [18], [19] have been introduced to bridge the gap between optimization

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and learning methods, effectively leveraging the observation model.

Despite the extensive work in the field of HMF, all current methods for addressing HMF issues have one prerequisite: the need for accurate registration of the two input images (HSI and MSI). The quality of the reconstructed high-resolution HSI depends largely on the accuracy of the registration [20], [21]. Even within the registration community, aligning HSI and MSI poses a challenging problem, as a single pixel in low-resolution HSI may cover hundreds of pixels in high-resolution MSI. Due to the significant spectral differences, spectral response functions (SRFs) and multiband images must be simultaneously considered during registration [20], [22].

This study introduces a novel fusion algorithm for HSI and MSI, the deep unfolding network enhanced by Transformer priors (DUNET), as illustrated in Fig. 1. The approach combines mutual information and deep priors to enhance the fusion performance. Initially, the method employs mutual information to align the HSI images. Throughout the end-to-end optimization process, we incorporate the observation matrix of HSIs, transforming the conventional iterative HSI super-resolution algorithm into a new nonregistered fusion approach. To better exploit the spatial–spectral information of HSIs, we also introduce a deep prior method that combines a mixed attention Transformer with a spatial-frequency module. Experimental results demonstrate significant improvements in this approach over mainstream algorithms on three commonly used HSI datasets.

The following summarizes the primary contributions of this article.

1) This article proposes a novel method called DUNET to tackle the simultaneous registration and fusion of HSI and MSI in satellite remote sensing. By jointly handling the registration and fusion tasks, the method applies consistency constraints more effectively, making the registration and fusion processes more coordinated and resulting in images with higher spatial and spectral consistency. Furthermore, the method decomposes the problem into multiple simple iterative steps, with clearly defined parameters and computational processes for each step, thereby enhancing the interpretability of the optimization process.
2) The proposed deep approximation operator incorporates HATs and spatial-frequency modules, which effectively leveraging the spatial–spectral information of HSI. This approach effectively leverages local details, global frequency information, and spectral channel correlations to offer a more efficient and precise hyperspectral and MSI fusion solution.
3) Extensive quantitative and visual experiments conducted on three benchmark HSI datasets have demonstrated the superior performance of DUNET compared with the existing mainstream algorithms. These experimental results fully validate the superiority of DUNET in HSI fusion and indicate its significant potential in addressing the registration and fusion issues of HSI and MSI.

The structure of the remaining sections of this article is as follows. Section II introduces the background relevant to our study, while Section III offers a comprehensive description of our proposed DUNET. This includes details on image registration methods, the rationale behind the model, network implementation, and proximity operations. Section IV presents the experimental results along with a detailed discussion, while Section V concludes the study and provides insights into future directions.

II. Related Works

This section summarizes notable recent research advances in HMF techniques, including registered and nonregistered HSI fusion methods.

A. Registered HSI Fusion

In recent years, significant strides have been achieved in the field of computer vision, driven by the rapid development and adoption of deep learning methodologies. However, directly applying deep learning for the fusion from LR-HSI and HR-MSI to HR-HSI often results in a process that lacks interpretability, akin to a black-box approach [23], [24], [25], [26], [27]. To address this issue, researchers have proposed deep unfolding-based HMF methods. For instance, Xie et al. [28] proposed an innovative approach to transform a low-rank HSI–MSI fusion model into a deep learning network, harnessing the advantages of deep learning for more precise and efficient HMF. Yang et al. [17] introduced a method named VaFuNet in their work, effectively leveraging multiscale nonlocal priors in HMF. Dong et al. [18] proposed an iterative HMF algorithm based on a deep HSI denoiser, incorporating domain knowledge likelihood and deep image priors. This method enhances the robustness and reliability of the fusion process, demonstrating high accuracy and robustness in handling both HMF tasks. In addition, a multidepth proximal learning network for HMF was introduced in [19], capable of adaptively inferring unknown spatial–spectral degradation and latent HR-HSI. Considering the data requirements of supervised methods, scholars have also proposed unsupervised methods based on deep priors. For example, Yang et al. [29] introduced a method called unsupervised deep tensor network (UDTN), aiming to enhance the resolution of HSI by fusing LR-HSI and HR-MSI.

In contrast, due to CNNs primarily capturing local information through convolutional operations, they may not perform as well as Transformers in capturing long-distance dependencies and global information. Despite its notable success in natural language processing (NLP) [30], the Transformer architecture encounters challenges in its direct application to image processing. To tackle this, researchers have proposed various Transformer networks for image-processing tasks. For instance, Jia et al. [31] fused HSI and MSI, leveraging the Transformer's ability to extract global information and self-supervised pretraining strategies, achieving outstanding fusion results. In addition, Deng et al. [32] utilized pyramid shuffle and reshuffle Transformers for HMF tasks, successfully addressing computational complexity and effectively handling information interaction among global blocks in remote sensing images. Other approaches include integrating a Transformer

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with 3-D convolutional neural networks (3D-CNNs) to learn priors of HSI [33], knowledge-driven deep unfolding [34], and a convolutional recurrent neural network with an embedded Transformer for HMF [35]. These methods have significantly improved performance in HMF tasks.

B. Nonregistered HSI Fusion

Over the past decade, the field of remote sensing image registration has witnessed the emergence of various methods [36], [37]. Traditional approaches typically involve iterative optimization, but recent research has introduced more innovative techniques. Zhou et al. [20] proposed a framework for registering high-spectral and high-resolution MSI by minimizing a least squares objective function while simultaneously considering spectral response and point spread functions, achieving high-precision registration applicable to diverse scenarios. Lin et al. [38] presented a method for HMF, leveraging spectral and spatial correlations between misaligned LR-HSI and HR-MSI. Fu et al. [39] addressed the challenges of simultaneous high-spectral image super-resolution and geometric alignment in mixed-camera systems, proposing a solution for misalignment. In addition, a novel algorithm for fusing misaligned HSI was introduced [22], employing registration and interpolation modeling to achieve superior fusion results. With the advancement of neural networks, unsupervised methods have gained attention. For instance, NonRegSRNet for nonrigid registration [40] and an unsupervised mutual Dirichlet network [41] have significantly enhanced image reconstruction accuracy and stability, making breakthroughs in HMF. This article [42] proposes a novel method for achieving unregistered HSI fusion by designing a registration–fusion consistency physical perception model to enhance the interpretability of the fusion results. This model can uniformly model the image registration and fusion processes, thereby improving the reliability and interpretability of the fusion results.

III. Methodology

A. Problem Formulation

We represent the potential HR-HSI as $\mathbf{Z} \in \mathbb{R}^{MN\times B}$, where B represents the number of spectral bands, and M and N represent height and width, respectively. The objective of HMF methods is to reconstruct the potential HR-HSI $\mathbf{Z}$ from the observed LR-HSI $\mathbf{X} \in \mathbb{R}^{mn\times B}$ and HR-MSI $\mathbf{Y} \in \mathbb{R}^{MN\times b}$, where $m \ll M$, $n \ll N$, and $b \ll B$. The relationship among $\mathbf{Z}$, LR-HSI, and HR-MSI can be formulated as a degradation model

$$\mathbf{X} = \mathbf{CZ} + \mathbf{N_1} \tag{1}$$

$$\mathbf{Y} = \mathbf{ZR} + \mathbf{N_2} \tag{2}$$

where $\mathbf{C} \in \mathbb{R}^{mn\times MN}$ and $\mathbf{R} \in \mathbb{R}^{B\times b}$ represent the degradation functions in the spatial and spectral domains, respectively. $\mathbf{N_1}$ and $\mathbf{N_2}$, respectively, represent the modeling errors (or noise) in the degradation relationships of the LR-HSI and the HR-MSI with the HR-HSI. Nomenclature provides an overview of all variable symbols employed in this article.

B. Image Registration Method

The mutual information image registration method [43] is a commonly used technique for image registration, which utilizes the information content in images to achieve alignment. First, the entropy and joint entropy of the overlapping parts of two images are calculated and then combined using the mutual information formula. Mathematically, mutual information can be expressed as follows:

$$\mathbf{MI}(\mathbf{X}, \mathbf{Y}) = \mathbf{H}(\mathbf{X}) + \mathbf{H}(\mathbf{Y}) - \mathbf{H}(\mathbf{X}, \mathbf{Y}) \tag{3}$$

where $\mathbf{H}(\mathbf{X})$ and $\mathbf{H}(\mathbf{Y})$ represent the entropy of variables $\mathbf{X}$ and $\mathbf{Y}$, respectively. $\mathbf{H}(\mathbf{X}, \mathbf{Y})$ represents the joint entropy of $\mathbf{X}$ and $\mathbf{Y}$. In order to effectively register two images, it is necessary to maximize the mutual information between the two images. From the above formula, it can be seen that in order to maximize the mutual information, it is required that the individual entropies $\mathbf{H}(\mathbf{X})$ and $\mathbf{H}(\mathbf{Y})$ are high, while the joint entropy $\mathbf{H}(\mathbf{X}, \mathbf{Y})$ is low. In the specific process of image registration, mutual information is used to estimate the deformation field from an HSI observed to an MSI. This process is represented as follows:

$$\omega_{\mathbf{X}\to\mathbf{Y}} = \mathbf{MI}(\mathbf{X}, \mathbf{Y}) \tag{4}$$

where the deformation field $\omega_{\mathbf{X}\to\mathbf{Y}}[i, j] = (\triangle x, \triangle y) \in \mathbb{R}^2$ represents the spatial transformation relationship of each pixel point from the HSI observed to the MSI, where i and j represent the horizontal and vertical coordinates of the pixel points in $\mathbf{X}$, and $\triangle x$ and $\triangle x$ represent the positional offset of the horizontal and vertical coordinates, respectively. By using the estimated deformation field, the HSI is resampled to obtain the registered HSI $\hat{\mathbf{X}}$. This process can be represented as follows:

$$\hat{\mathbf{X}} = \mathbf{warp}(\mathbf{X}, \omega_{\mathbf{X}\to\mathbf{Y}}) \tag{5}$$

where $\mathbf{warp}(\cdot)$ represents the resampling function.

C. Modeling Motivation

The symbols $\mathbf{X}$, $\hat{\mathbf{X}}$, and $\mathbf{Y}$ represent the unregistered LR-HSI image, the registered image, and the HR-MSI image, respectively. These are derived from (1) and (2)

$$\begin{aligned} \hat{\mathbf{X}} &= \mathbf{CZ} \\ \mathbf{Y} &= \mathbf{ZR} \\ \hat{\mathbf{X}} &= \mathbf{P}(\mathbf{X}) \end{aligned} \tag{6}$$

where $\mathbf{P}(\cdot)$ represents the transformation process of aligning the LR-HSI $\mathbf{X}$ to the reference image $\hat{\mathbf{X}}$ using mutual information.

To mitigate overfitting, additional constraints need to be introduced. When the low-resolution observed HSI $\mathbf{X}$ is registered to an ideal state, meaning the registered image $\hat{\mathbf{X}}$ achieves optimal similarity with the MSI $\mathbf{Y}$, the spectral degradation result $\hat{\mathbf{X}}\mathbf{R}$ will be equal to the spatial degradation result $\mathbf{CY}$ of $\mathbf{Y}$, as shown below

$$\hat{\mathbf{X}}\mathbf{R} = \mathbf{CY}. \tag{7}$$

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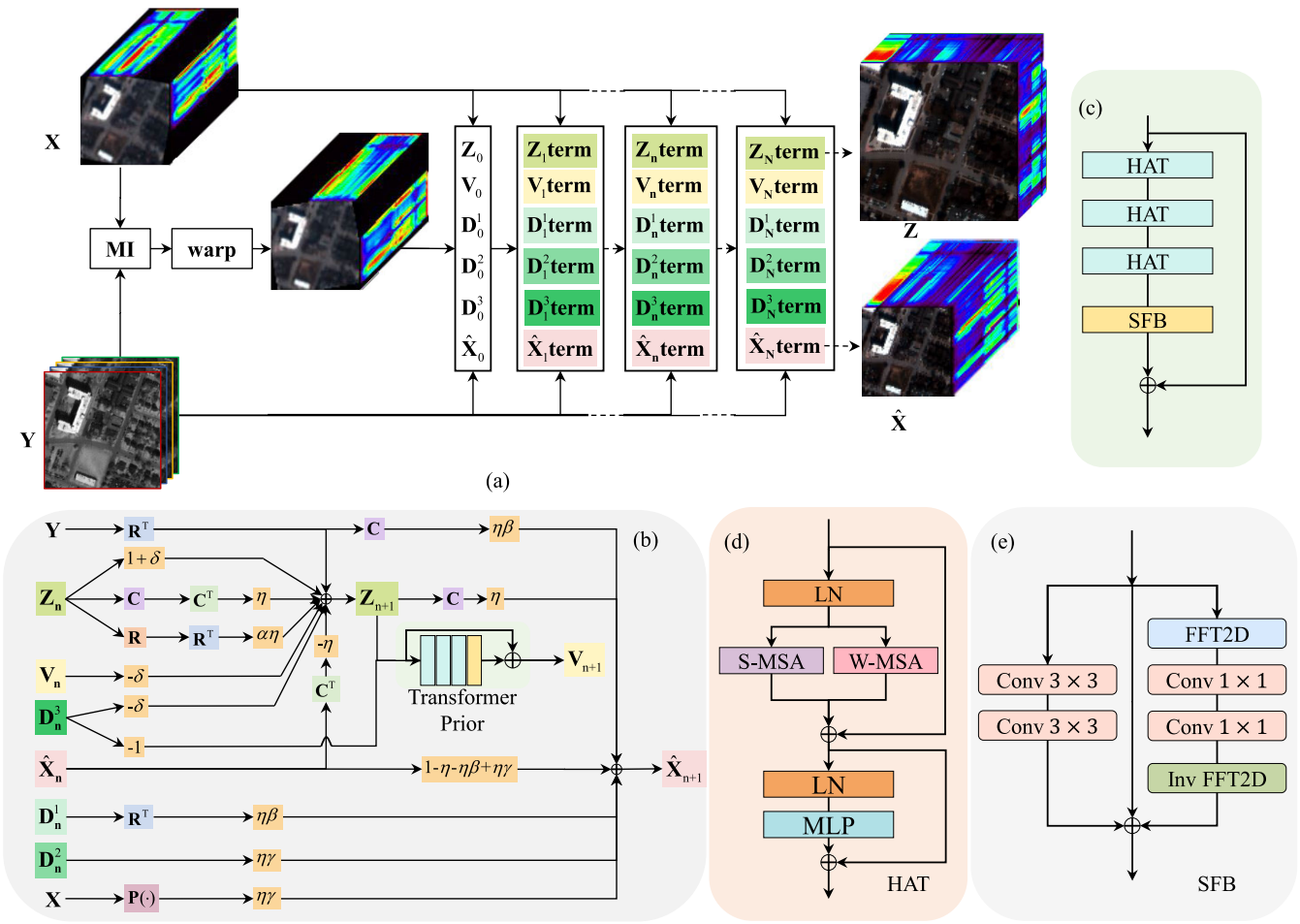


Fig. 1. (a) Overall flowchart of the proposed method. Initially, LR-HSI $\mathbf{X}$ and HR-MSI $\mathbf{Y}$ are registered through mutual information. Subsequently, utilizing the registered HSI along with the unregistered HSI and the joint equation of HR-MSI, this process is learned through a deep unfolding network. (b) Specific details of the deep unfolding network. (c) HAT frequency approximation. (d) HAT. (e) SFB.

Combining (6) and (7), the optimization objective can be expressed as follows:

$$\min_{\mathbf{Z},\hat{\mathbf{X}}} \|\hat{\mathbf{X}} - \mathbf{CZ}\|_F^2 + \alpha\|\mathbf{Y} - \mathbf{ZR}\|_F^2 + \lambda J(\mathbf{Z})$$
$$\text{s.t.} \hat{\mathbf{X}}\mathbf{R} = \mathbf{CY}, \hat{\mathbf{X}} = \mathbf{P}(\mathbf{X}). \tag{8}$$

By imposing such constraints on $\mathbf{P}(\cdot)$, it is possible to effectively reduce the solution space for registration, enabling the proposed framework to appropriately eliminate unknown misalignments between observed images, thereby achieving improved HMF performance.

D. Model Optimization

For the proposed DUNET method, there is an optimization (8), and ADMM can be used to solve it. By introducing three variables $\mathbf{V} \in \mathbb{R}^{MN\times B}$, the optimization (8) can be rewritten as follows:

$$\min_{\mathbf{Z},\hat{\mathbf{X}},\mathbf{V}} f(\mathbf{Z}, \hat{\mathbf{X}}, \mathbf{V})$$
$$\text{s.t.} \hat{\mathbf{X}}\mathbf{R} = \mathbf{CY}, \hat{\mathbf{X}} = \mathbf{P}(\mathbf{X}), \mathbf{Z} = \mathbf{V} \tag{9}$$

where

$$f(\mathbf{Z}, \hat{\mathbf{X}}, \mathbf{V}) = \|\hat{\mathbf{X}} - \mathbf{CZ}\|_F^2 + \alpha\|\mathbf{Y} - \mathbf{ZR}\|_F^2 + \lambda J(\mathbf{V}). \tag{10}$$

The augmented Lagrangian function of (9) can be written as follows:

$$\mathcal{L}(\mathbf{Z}, \hat{\mathbf{X}}, \mathbf{V}, \mathbf{D}^1, \mathbf{D}^2, \mathbf{D}^3) = f(\mathbf{Z}, \hat{\mathbf{X}}, \mathbf{V}) + \beta\|\hat{\mathbf{X}}\mathbf{R} - \mathbf{CY} - \mathbf{D}^1\|_F^2 + \gamma\|\hat{\mathbf{X}} - \mathbf{P}(\mathbf{X}) - \mathbf{D}^2\|_F^2 + \delta\|\mathbf{Z} - \mathbf{V} - \mathbf{D}^3\|_F^2 \tag{11}$$

where α, β, γ, and δ are hyperparameters. The optimization problem of $\mathcal{L}$ with respect to each variable is as follows.

1) $\mathbf{Z}$ and $\hat{\mathbf{X}}$ Subproblems: Since we are only dealing with the problems of $\mathbf{Z}$ and $\hat{\mathbf{X}}$, we can express (11) as follows:

$$\min_{\mathbf{Z},\hat{\mathbf{X}}} \mathcal{L}(\mathbf{Z}, \hat{\mathbf{X}}) = \|\hat{\mathbf{X}} - \mathbf{CZ}\|_F^2 + \alpha\|\mathbf{Y} - \mathbf{ZR}\|_F^2 + \beta\|\hat{\mathbf{X}}\mathbf{R} - \mathbf{CY} - \mathbf{D}^1\|_F^2 + \gamma\|\hat{\mathbf{X}} - \mathbf{P}(\mathbf{X}) - \mathbf{D}^2\|_F^2 + \delta\|\mathbf{Z} - \mathbf{V} - \mathbf{D}^3\|_F^2. \tag{12}$$

We can use the gradient descent method to sequentially update two target variables, and the gradient calculation for each variable is as follows:

$$\frac{\partial \mathbf{L}}{\partial \mathbf{Z}} = \mathbf{C}^T(\hat{\mathbf{X}} - \mathbf{CZ}) + \alpha(\mathbf{Y} - \mathbf{ZR})\mathbf{R}^T + \delta(\mathbf{Z} - \mathbf{V} - \mathbf{D}^3) \tag{13}$$

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$$\frac{\partial \mathbf{L}}{\partial \hat{\mathbf{X}}} = (\hat{\mathbf{X}} - \mathbf{CZ}) + \beta(\hat{\mathbf{X}}\mathbf{R} - \mathbf{CY} - \mathbf{D}^1)\mathbf{R}^T + \gamma(\hat{\mathbf{X}} - \mathbf{P}(\mathbf{X}) - \mathbf{D}^2). \tag{14}$$

Here, the nth iteration of model optimization can be represented as follows:

$$\mathbf{Z}_{n+1} = \mathbf{Z}_n - \eta(\mathbf{C}^T(\hat{\mathbf{X}}_n - \mathbf{CZ}_n) + \alpha(\mathbf{Y} - \mathbf{Z}_n\mathbf{R})\mathbf{R}^T + \delta(\mathbf{Z}_n - \mathbf{V}_n - \mathbf{D}_n^3)) \tag{15}$$

$$\hat{\mathbf{X}}_{n+1} = \hat{\mathbf{X}}_n - \eta((\hat{\mathbf{X}}_n - \mathbf{CZ}_{n+1}) + \beta(\hat{\mathbf{X}}_n\mathbf{R} - \mathbf{CY} - \mathbf{D}_n^1)\mathbf{R}^T + \gamma(\hat{\mathbf{X}}_n - \mathbf{P}(\mathbf{X}) - \mathbf{D}_n^2)) \tag{16}$$

where the parameter η represents the step size for each iteration and the subscript indicates the stage of iteration.

2) $\mathbf{V}$ Subproblem: Since we are only dealing with the problems of $\mathbf{V}$, we can express (11) as follows:

$$\min_{\mathbf{V}} \mathcal{L}(\mathbf{V}) = \|\mathbf{Z} - \mathbf{V} - \mathbf{D}^3\|_F^2 + \lambda J(\mathbf{V}). \tag{17}$$

The V subproblem can alternatively be expressed as follows:

$$\mathbf{V}_{n+1} = \text{prox}_{\lambda J}(\mathbf{Z}_{n+1} - \mathbf{D}_n^3) \tag{18}$$

denoting an approximate operator of $\lambda J(\cdot)$ at point $\mathbf{Z}_{n+1} - \mathbf{D}_n^3$. As there is no explicit mathematical expression for the deep prior, finding a closed-form solution for the V subproblem becomes difficult.

3) Updating Multipliers: The multipliers associated with L is updated as follows:

$$\begin{aligned} \mathbf{D}_{n+1}^1 &= \mathbf{D}_n^1 - (\hat{\mathbf{X}}_{n+1}\mathbf{R} - \mathbf{CY}) \\ \mathbf{D}_{n+1}^2 &= \mathbf{D}_n^2 - (\hat{\mathbf{X}}_{n+1} - \mathbf{P}(\mathbf{X})) \\ \mathbf{D}_{n+1}^3 &= \mathbf{D}_n^3 - (\mathbf{Z}_{n+1} - \mathbf{V}_{n+1}). \end{aligned} \tag{19}$$

E. Network Design

The article delves into the iterative process of addressing the HR-HSI reconstruction problem until the subproblems involving $\mathbf{Z}$, $\mathbf{V}$, $\mathbf{D}^1$, $\mathbf{D}^2$, $\mathbf{D}^3$, and $\hat{\mathbf{X}}$ converge to local minima. While this iterative approach shows promise in theory, its practical implementation requires significant time and computational resources. In response to this challenge, we present DUNET, a method that unfolds the iterative process and utilizes deep learning networks to perform each iteration step. This approach greatly improves computational efficiency. At the core of DUNET is its forward computation, which facilitates HR-HSI inference while implicitly encapsulating the degradation model and data priors. Via this computation, DUNET progressively reconstructs high-resolution spectral images, thereby enhancing both the accuracy and robustness of inference. Moreover, DUNET embraces an end-to-end methodology, optimizing the complete reconstruction process within a unified framework. The strength of this approach lies in its capacity to autonomously learn deep priors, degradation model parameters, and hyperparameters without requiring manual adjustments or predefined specifications. This capability improves the model's capacity for generalization and its suitability for diverse scenarios.

1) Iterations Unfolding: In order to streamline network design, both HSI and MSI are portrayed in tensor format rather than matrix format within this section. This representation enables the model to more efficiently manage high-dimensional data structures, thereby improving the flexibility and performance of the network. Considering an unregistered LR-HSI denoted as $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{m\times n\times B}$ and an HR-MSI denoted as $\boldsymbol{\mathcal{Y}} \in \mathbb{R}^{M\times N\times b}$, the $\mathbf{Z}$ and $\hat{\mathbf{X}}$ subproblems in (15) and (16) can be reformulated in tensor notation as follows:

$$\boldsymbol{\mathcal{Z}}_{n+1} = \boldsymbol{\mathcal{Z}}_n - \eta(\boldsymbol{\mathcal{C}}^T(\hat{\boldsymbol{\mathcal{X}}}_n - \boldsymbol{\mathcal{C}}\boldsymbol{\mathcal{Z}}_n) + \alpha(\boldsymbol{\mathcal{Y}} - \boldsymbol{\mathcal{Z}}_n \times_3 \mathbf{R}) \times_3 \mathbf{R}^T + \delta(\boldsymbol{\mathcal{Z}}_n - \boldsymbol{\mathcal{V}}_n - \boldsymbol{\mathcal{D}}_n^3)) \tag{20}$$

$$\hat{\boldsymbol{\mathcal{X}}}_{n+1} = \hat{\boldsymbol{\mathcal{X}}}_n - \eta((\hat{\boldsymbol{\mathcal{X}}}_n - \boldsymbol{\mathcal{C}}\boldsymbol{\mathcal{Z}}_{n+1}) + \beta(\hat{\boldsymbol{\mathcal{X}}}_n \times_3 \mathbf{R} - \boldsymbol{\mathcal{C}}\boldsymbol{\mathcal{Y}} - \boldsymbol{\mathcal{D}}_n^1) \times_3 \mathbf{R}^T + \gamma(\hat{\boldsymbol{\mathcal{X}}}_n - \mathbf{P}(\boldsymbol{\mathcal{X}}) - \boldsymbol{\mathcal{D}}_n^2)) \tag{21}$$

where $\boldsymbol{\mathcal{Z}}_n \in \mathbb{R}^{M\times N\times B}$ and $\hat{\boldsymbol{\mathcal{X}}}_n \in \mathbb{R}^{m\times n\times B}$ represent the intermediate HR-HSI and aligned LR-HSI at the nth iterative phase, respectively. $\times_3$ denotes tensor multiplication in the third mode. In this context, we introduce symbols $\boldsymbol{\mathcal{C}}$ and $\boldsymbol{\mathcal{C}}^T$ to represent spatial degradation operations and their reversals on tensors, akin to those delineated in (15), and we approximate these operations using convolutional and transposed convolutional layers with kernel sizes and strides set according to the upsampling factor r. In addition, spectral degradation can be executed by employing a 1×1 convolutional layer with b feature maps to carry out mode-3 tensor multiplication with $\mathbf{R}^T$. Similarly, mode-3 tensor multiplication with $\mathbf{R}$ can be accomplished using a 1×1 convolutional layer with B feature maps.

At the nth iteration phase, the V subproblem can also be expressed in tensor format as follows:

$$\boldsymbol{\mathcal{V}}_{n+1} = \text{prox}_{\lambda J}(\boldsymbol{\mathcal{Z}}_{n+1} - \boldsymbol{\mathcal{D}}_n^3) \tag{22}$$

where $\boldsymbol{\mathcal{V}} \in \mathbb{R}^{M\times N\times B}$ represents the auxiliary variable. In Section III-F, we will provide a comprehensive overview of the network architecture for the proximity operator.

In tensor format, the multipliers subproblem at the nth iteration phase can be reformulated as follows:

$$\begin{aligned} \boldsymbol{\mathcal{D}}_{n+1}^1 &= \boldsymbol{\mathcal{D}}_n^1 - (\hat{\boldsymbol{\mathcal{X}}}_{n+1} \times_3 \mathbf{R} - \boldsymbol{\mathcal{C}}\boldsymbol{\mathcal{Y}}) \\ \boldsymbol{\mathcal{D}}_{n+1}^2 &= \boldsymbol{\mathcal{D}}_n^2 - (\hat{\boldsymbol{\mathcal{X}}}_{n+1} - \mathbf{P}(\boldsymbol{\mathcal{X}})) \\ \boldsymbol{\mathcal{D}}_{n+1}^3 &= \boldsymbol{\mathcal{D}}_n^3 - (\boldsymbol{\mathcal{Z}}_{n+1} - \boldsymbol{\mathcal{V}}_{n+1}). \end{aligned} \tag{23}$$

2) Iterations Initialization: During the initial iteration phase, the estimated HR-HSI is initialized using the spectrally upsampled $\boldsymbol{\mathcal{Y}}$. The estimation of the low-resolution HSI is obtained by registering $\boldsymbol{\mathcal{X}}$ with $\boldsymbol{\mathcal{Y}}$ using the mutual information method, which is formulated as follows:

$$\begin{cases} \boldsymbol{\mathcal{Z}}_0 &= \boldsymbol{\mathcal{Y}} \times_3 \mathbf{R} \\ \hat{\boldsymbol{\mathcal{X}}}_0 &= \mathbf{P}(\boldsymbol{\mathcal{X}}) \end{cases} \tag{24}$$

where $\boldsymbol{\mathcal{Z}}_0 \in \mathbb{R}^{M\times N\times B}$ is the initialized HR-HSI and $\hat{\boldsymbol{\mathcal{X}}}_0$ represents the initialized registered LR-HSI.

We derive the DUNET by expanding (20)–(24), resulting in the creation of several modules corresponding to the iterative

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phases. These modules execute the $\mathcal{Z}$, $\mathcal{V}$, $\mathcal{D}^1$, $\mathcal{D}^2$, $\mathcal{D}^3$, and $\hat{\mathcal{X}}$ subproblems, as illustrated in Fig. 1. Consisting of degradation parameters (e.g., $\mathcal{C}$ and $\mathbf{R}$), the deep prior proximity and hyperparameters (e.g., α, β, γ, and δ) described in (20)–(24) are the network parameters. Below is a summary of the overall algorithm for HMF, as presented in Algorithm 1.

Algorithm 1 Proposed DUNET

Input: unregistered LR-HSI $\mathcal{X}$, HR-MSI $\mathcal{Y}$;
Initialize: $\mathcal{Z}_0$ and $\hat{\mathcal{X}}_0$ via Eq. (24);$\mathcal{V}_0$, $\mathcal{D}_0^1$, $\mathcal{D}_0^2$, $\mathcal{D}_0^3$ by zero matrix.
while $n \leq N$ **do**
    Update $\mathcal{Z}_{n+1}$ via Eq. (20);
    Update $\hat{\mathcal{X}}_{n+1}$ via Eq. (21);
    Update $\mathcal{V}_{n+1}$ via Eq. (22);
    Update $\mathcal{D}_{n+1}^1$, $\mathcal{D}_{n+1}^2$, $\mathcal{D}_{n+1}^3$ via Eq. (23);
end while
Output: $\mathcal{Z}_N$, $\hat{\mathcal{X}}_N$

F. Proximity Operator

By unfolding the iterations of Algorithm 1 into a deep learning network, we propose DUNET, as shown in Fig. 1. Multiple stages in DUNET correspond to the iterative process outlined in Algorithm 1. Each stage involves solving three implicit variables, namely, $\mathcal{Z}_n$, $\hat{\mathcal{X}}_n$, and $\mathcal{V}_n$. The auxiliary variables $\mathcal{Z}_n$ and $\hat{\mathcal{X}}_n$ are solved according to 20 and 21, respectively. In this section, the variable $\mathcal{V}_n$ is solved utilizing a deep proximal operator. To better describe this operator, we first discuss three modules: the HAT and the SFB.

1) Hybrid Attention Transformer: In spectral self-attention (S-MSA), each pixel incorporates global information to activate corresponding pixels based on channel attention weights, enhancing the model's ability to capture global structure and features and improving performance and generalization. In addition, studies have shown that convolution aids Transformers in obtaining better visual representations or achieving easier optimization. As a result, S-MSA is integrated into the window Transformer block [32], [44] to bolster the network's representation capability. In the window Transformer block, we introduce an S-MSA block after the first LayerNorm (LN) layer, running it in parallel with the window-based multihead self-attention (W-MSA) module, as depicted in Fig. 1(d). To balance W-MSA and S-MSA, we multiply the output of S-MSA by a small constant μ. Overall, for a given input feature $\mathcal{F}_0$, the entire process of HAT can be described as follows:

$$\begin{aligned}\mathcal{M}_l &= \mathrm{LN}(\mathcal{F}_{l-1})\\ \mathcal{N}_l &= \text{W-MSA}(\mathcal{M}_l) + \mu\text{S-MSA}(\mathcal{M}_l) + \mathcal{F}_{l-1}\\ \mathcal{F}_l &= \mathrm{MLP}(\mathrm{LN}(\mathcal{N}_l)) + \mathcal{N}_l\end{aligned} \tag{25}$$

where $\mathcal{M}_l$ and $\mathcal{N}_l$ represent intermediate features, while $\mathcal{F}_L$ represents the output of HAT, $l \in \{1, 2, \ldots, L\}$. It is worth noting that we treat each pixel as an embedded token, implying that we set the patch size to 1, as per the configuration in [45]. MLP stands for multilayer perceptron.

a) Windowed Self-Attention: To compute the self-attention module, we start by partitioning the input feature into (HW/M^2) local windows, each of size $M \times M$, and then compute self-attention within each window. For a local window feature $\mathbf{F}_w \in \mathbb{R}^{M^2 \times X}$, we compute query, key, and value matrices through linear mappings, denoted as $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$, respectively. Subsequently, we express the W-MSA as follows:

$$\text{W-Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}(\mathbf{Q}\mathbf{K}^T/\sqrt{d})\mathbf{V} \tag{26}$$

where d represents the dimensionality of queries/keys.

b) Spectral Self-Attention: Within the Transformer block, three 1×1 convolutional layers are first applied to $\mathcal{F}_0 \in \mathbb{R}^{W \times H \times C}$, followed by reshaping to generate the query (Q), key (K), and value (V) matrices: $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{WH \times C'}$, where C' is the number of channels in the 1×1 convolutional layers. The S-MSA refines the feature embedding according to the following:

$$\text{S-Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathbf{V} \cdot \mathrm{softmax}(\mathbf{K}^T\mathbf{Q}/w) \tag{27}$$

where w serves as a trainable scaling parameter employed to regulate the softmax operation.

2) Spatial-Frequency Block: To enhance the representation capacity of the Transformer and make the frequency information [46], [47] of the image more explicit, we designed the SFB. As shown in Fig. 1(e), the SFB network consists of frequency and spatial branches, each receiving the input feature $\mathcal{F}_L$ and producing corresponding frequency and spatial features. In the following, we will individually introduce these two branches.

In the frequency branch, the frequency network is devised to extract features enhanced through frequency manipulation techniques. Here are the details of feature acquisition

$$\mathrm{FT}(\mathcal{F}_L) = \hat{F}_{\mathrm{T}}(C_1(C_1(F_T(\mathcal{F}_L)))) \tag{28}$$

where C_1 denotes a 1×1 convolutional layer, $\mathrm{F_T}$ represents the fast Fourier transform (FFT) layer, and $\hat{F}_{\mathrm{T}}$ represents the inverse FFT layer, and then transforming it into the frequency domain using 2-D FFT to extract global information for generating high-quality frequency features. Finally, the inverse 2-D FFT operation to convert the frequency features back to spatial features.

Furthermore, exploration of spatial information is required. We construct the spatial branch using convolutional layers and activation functions to enhance the expressive power of features, thereby obtaining refined spatial features

$$S(\mathcal{F}_L) = C_3(C_3(\mathcal{F}_L)) + \mathcal{F}_L \tag{29}$$

where C_3 denotes a 3×3 convolutional layer.

Based on the frequency branch and spatial branch, the output of SFB is represented as follows:

$$\mathrm{SFB}(\mathcal{F}_L) = \mathrm{FT}(\mathcal{F}_L) + S(\mathcal{F}_L). \tag{30}$$

In this particular scenario, the proximity for the $\mathcal{V}$ subproblem at the nth iteration phase can be formulated as follows:

$$\mathcal{V}_{n+1} = \mathcal{F}_0 + \mathrm{SFB}(\mathcal{F}_L). \tag{31}$$

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G. Implementation Details of Training

All parameters within DUNET undergo end-to-end training. The loss function is constructed to include the $L1$ norms of the reconstruction errors for both the latent HR-HSI $\mathcal{Z}$ and the latent LR-HSI $\hat{\boldsymbol{X}}$. In addition, the Fourier loss of the latent HR-HSI is incorporated into the training process. The process is shown as follows:

$$\begin{aligned}\text{loss} = &\frac{1}{T}\sum_{t}\|\mathcal{Z}^{(t)} - \tilde{\mathcal{Z}}^{(t)}\|_1 \\ &+ \frac{\tau_1}{T}\sum_{t}\|\mathcal{FFT}(\mathcal{Z}^{(t)}) - \mathcal{FFT}(\tilde{\mathcal{Z}}^{(t)})\|_1 \\ &+ \tau_2\|\hat{\boldsymbol{X}}^{(t)} - \tilde{\boldsymbol{X}}^{(t)}\|_F^2 \end{aligned} \tag{32}$$

where $\mathcal{Z}^{(t)}$ and $\hat{\boldsymbol{X}}^{(t)}$ are the tth ground truths of HR-HSI and LR-HSI, respectively. $\tilde{\mathcal{Z}}^{(t)}$ and $\tilde{\boldsymbol{X}}^{(t)}$ are their estimations inferred by DUNET. $\mathcal{FFT}$ represents the FFT operation. $t = 1, 2, \ldots, T$, where T is the number of training samples. τ_1 and τ_2 modulate the significance of FFT loss and LR-HSI, while the adaptive moment estimation (ADAM) optimizer [48] dynamically adjusts network parameters to minimize the loss function.

It should be noted that to train the DL network, a sufficient number of paired image patches $\{\mathcal{Z}^{(t)}, (\mathcal{X}^{(t)}, \mathcal{Y}^{(t)})\}$ need to be extracted as training samples. However, in real-world applications, the ground-truth HR-HSI is often unavailable, making it unrealistic to directly extract $\mathcal{Z}^{(t)}$. To overcome this challenge, we adopt the strategy proposed in [28], which follows the Wald protocol [49] for creating training samples. By training a deep learning network, we can apply it to real LR-HSI and HR-MSI images to generate corresponding HR-HSI images.

IV. Experiments

In this section, we provide a detailed exposition of the experimental setup and network performance evaluation for a deep prior-based unregistered HMF network aimed at addressing the HMF problem. Initially, we introduce the datasets and evaluation metrics employed in the experiments. Subsequently, we benchmark our proposed DUNET against state-of-the-art HMF methodologies. We compare DUNET with three classical methods (coupled nonnegative matrix factorization (CNMF) [50],[1] the subspace regularized method (HySure) [51],[2] and coupled spectral unmixing (CSU) [4][3]), along with five state-of-the-art techniques (current approaches), i.e., unregistered and unsupervised mutual Dirichlet-Net (u^2-MDN) [41],[4] unaligned HSI fusion via registration and interpolation modeling (UHIF-RIM) [22],[5] a novel model-guided deep convolutional network (MoG-DCN) [18],[6] a novel multiscale spatial–spectral Transformer network (MSST-Net) [31],[7] and a novel pyramid shuffle-and-reshuffle Transformer (PSRT) [32].[8] The experiments were conducted using PyTorch on an NVIDIA 2080TI RTX computer.

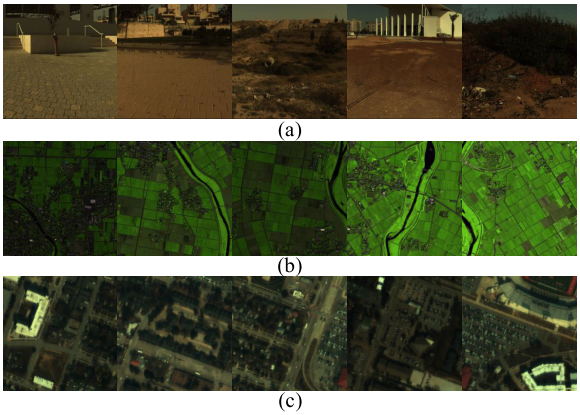


Fig. 2. Examples of the testing images selected from (a) ICVL, (b) Chikusei, and (c) Houston datasets.

A. Experimental Dataset

The effectiveness of the proposed approach is assessed using three publicly available simulated HSI datasets: the interdisciplinary computational vision lab (ICVL) dataset,[9] the Chikusei dataset,[10] and the Houston dataset.[11] Partial images from the test sets of these three datasets are depicted in Fig. 2.

1) ICVL: The first dataset is the ICVL HSI dataset, captured using a Specim PS Kappa DX4 hyperspectral camera. Each HSI in this dataset has a spatial size of 1392 × 1300 pixels. It consists of 31 spectral bands covering the range of 400–700 nm.

2) Chikusei: This dataset comprises aerial hyperspectral images taken over agricultural and urban areas of Tsukijo, Ibaraki Prefecture, Japan, using visible and near-infrared (NIR) imaging sensors. The dataset includes 128 bands spanning from 363 to 1018 nm in the spectral domain, with scenes comprising 2517 × 2335 pixels. Following the elimination of black edges in the spatial domain, a central area measuring 2048 × 2048 pixels was cropped and utilized for our experiments.

3) Houston: The Houston 2018 dataset, sourced from the 2018 IEEE GRSS Data Fusion Competition, comprises hyperspectral data acquired over the University of Houston campus in February 2017. This dataset comprises spatial data captured by the ITRES CASI-1500 sensor, with the dimensions of 601 × 2384 pixels, along with LiDAR data acquired from the Optech Titam multiwave (MW) (14SEN/CON340) sensor. The hyperspectral dataset comprises 50 spectral bands ranging from 380 to 1050 nm.

The original hyperspectral images serve as ground-truth data, from which LR-HSI and HR-MSI are simulated through spatial and spectral degradation. Specifically, the original images undergo processes, such as downsampling and spectral

[1] http://naotoyokoya.com/Download.html
[2] https://github.com/alfaiate/HySure
[3] https://github.com/lanha/SupResPALM
[4] https://github.com/yingutk/u2MDN
[5] https://github.com/YingJiacheng/UHIF-RIM
[6] https://github.com/chengerr/Model-Guided-Deep-Hyperspectral-Image-Super-resolution
[7] https://github.com/jx-mzc/MSST-Net
[8] https://github.com/Dengshangqi/PSRT
[9] http://icvl.cs.bgu.ac.il/hyperspectral/
[10] http://naotoyokoya.com/Download.html
[11] https://hyperspectral.ee.uh.edu/page_id=1075

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filtering, to simulate observations at different resolutions and spectral ranges. These simulated LR-HSI and HR-MSI are then used to evaluate the performance of the proposed method. LR-HSI is obtained by downsampling the reference image every r pixels in both the vertical and horizontal directions using a Gaussian filter (with mean 0 and standard deviation 2), corresponding to an extraction factor of $r \times r$ (r is set to 4). To simulate misregistration, each LR-HSI is rotated by 2° and 5°. The subspectral sampling matrix $\mathbf{R}$, adopting the response function of the Nikon D700 camera,[12] is utilized to spectrally downsample the HR-HSI, thereby simulating the HR-MSI of the same scene.

B. Quantitative Indicators

The evaluation criteria utilized in this study encompass a spectrum of spatial and spectral quality assessments. Larger values indicate better restoration outcomes for HSI, as seen in metrics, such as mean peak signal-to-noise ratio (MPSNR), universal image quality index (UIQI), and mean structural similarity index (MSSIM). Conversely, smaller values signify superior restoration results, as evidenced by metrics, such as spectral angle mapping (SAM), erreur relative global adimensionnelle synthese (ERGAS), and root-mean-square error (RMSE).

C. Experimental Results

The evaluation outcomes of different methodologies are delineated in Tables I–III. Each dataset, namely, ICVL, Chikusei, and Houston, underwent assessment employing distinct fusion methodologies across 20, 8, and 8 test datasets, correspondingly. Subsequently, the mean evaluation metrics were computed for each dataset. According to Tables I–III, the proposed method performs best in terms of MPSNR, UIQI, and MSSIM metrics and achieves the lowest values in terms of RMSE, ERGAS, and SAM metrics, indicating that the proposed method reconstructs HSI with better spatial structures and smaller spectral distortions compared with the reference methods.

1) Results on ICVL Dataset: The fusion results of the ICVL dataset presented in Fig. 3 indicate that the HySure method yields darker experimental outcomes, while the CNMF and MoG-DCN methods produce black areas at the four corners. Other methods produce similar fusion outcomes. To visually illustrate discrepancies, this study generates heat maps of mean-squared error images in Fig. 3. In these maps, shades of blue indicate minor errors, while shades of red represent significant errors. As shown in Fig. 4, CNMF, HySure, CSU, u^2MDN, UHIF-RIM, MoG-DCN, and MoE-PNP methods exhibit noticeable reconstruction errors. The MSST and PSRT methods display prominent reconstruction errors along the edges of buildings. Our proposed method demonstrates the least reconstruction errors. Fig. 5(a) and (b) displays per-band PSNR and SSIM values for the reference and fused images, respectively. The presented values indicate that the proposed method consistently achieves the highest indices for each band, emphasizing its superior performance characterized by minimal reconstruction errors. Moreover, Fig. 5(c) demonstrates the SAM values across each band for both the reference and fused images.

2) Results on Chikusei Dataset: Fig. 6 illustrates the fusion outcomes of the Chikusei dataset, indicating that the HySure method results in a brighter image compared with other techniques. As depicted in Fig. 7, significant reconstruction errors are noticeable in methods, such as CNMF, HySure, CSU, and UHIF-RIM. MoG-DCN and PSRT methods display evident reconstruction errors along the boundaries of fields and water channels, while the u^2MDN, MSST, and MoE-PNP methods exhibit some reconstruction errors around building areas. In contrast, our proposed approach showcases the minimal reconstruction errors. Fig. 8(a) and (b) displays the per-band PSNR and SSIM values for the reference and fused images, respectively. These figures highlight that our proposed method attains the highest indices across all bands, emphasizing its outstanding performance even with minimal reconstruction errors. Furthermore, Fig. 8(c) illustrates the SAM values across each band for both the reference and fused images.

3) Results on Houston Dataset: Fig. 9 depicts the fusion outcomes of the Houston dataset, showing that compared with other techniques, the HySure and PSRT methods result in brighter images, while the CSU method leads to significant loss of spectral information. As illustrated in Fig. 10, notable reconstruction errors are observed in methods, such as CNMF, HySure, CSU, u^2MDN, and UHIF-RIM. MoG-DCN, PSRT, and MoE-PNP methods exhibit noticeable reconstruction errors around white buildings, while the MSST method introduces grid-like noise. In contrast, our proposed method demonstrates minimal reconstruction errors. Fig. 11(a) and (b) displays the per-band PSNR and SSIM values for the reference and fused images, respectively. These results underscore that our proposed method achieves the highest indices across all bands, indicating its exceptional performance despite minimal reconstruction errors. In addition, Fig. 11(c) illustrates the SAM values across each band for the reference and fused images.

4) Visualization of Different Stages: Utilizing the test image from the Houston dataset (as depicted in Fig. 9), we exemplify the progression of intermediate HSI $\mathcal{Z}_n$ generated by DUNET at various stages. In Fig. 12, the inferred HSI $\mathcal{Z}_n$ from DUNET at different stages is presented. It can be seen from the figures that as the stages progress, the fusion results become clearer.

5) Results on LR-HSI: From Table IV, it can be seen that the PSNR and SSIM values of the LR-HSI generated by our proposed DUNET method are higher compared with those generated by the original MI [52] and NED [22] methods. As shown in Fig. 13, our method for generating LR-HSI utilizes the HR-MSI to fill in the black regions left due to rotation, which were present in the LR-HSI generated by MI and NED. Therefore, we believe that the DUNET method is effective.

In conclusion, the proposed supervised unregistered HSI fusion method demonstrates superior performance across various metrics, as well as in the color and error images, for

[12]https://www.maxmax.com/spectral_response.htm

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TABLE I

EVALUATION OF THE PROPOSED AND COMPARATIVE TECHNIQUES ON THE ICVL DATASET ENTAILED UTILIZING A GAUSSIAN BLUR KERNEL AND A SCALING FACTOR OF 4. PRESENTED ARE THE MEAN MPSNR, RMSE, ERGAS, SAM, UIQI, AND MSSIM VALUES, WITH ↑ INDICATING SUPERIOR PERFORMANCE WITH LARGER VALUES, AND ↓ SUGGESTING IMPROVED PERFORMANCE WITH SMALLER VALUES. THE TOP-RANKED AND SECOND-BEST OUTCOMES ARE **EMPHASIZED** IN BOLD AND UNDERLINED, RESPECTIVELY

angle	Indices	CNMF [50]	HySure [51]	CSU [4]	u^2MDN [41]	UHIF-RIM [22]	MoG-DCN [18]	MSST [31]	PSRT [32]	MoE-PNP [42]	DUNET
2°	MPSNR↑	29.9712	29.2191	31.4045	43.1519	27.9257	31.1808	45.3817	37.5326	34.9307	**48.8147**
	RMSE↓	9.7293	11.8232	8.3583	4.6071	11.5067	8.4133	1.5260	4.2595	5.5721	**1.0786**
	ERGAS↓	5.2226	5.7068	4.9828	1.4357	3.9159	2.3506	0.6658	1.6937	2.6494	**0.5562**
	SAM↓	2.8256	7.1682	3.1047	6.4351	4.8626	3.8578	1.8717	1.7085	5.1994	**1.5552**
	UIQI↑	0.7201	0.7737	0.8006	0.8896	0.5710	0.8077	0.9432	0.9442	0.9084	**0.9502**
	MSSIM↑	0.9068	0.8375	0.9469	0.9432	0.6578	0.9495	0.9936	0.9899	0.9757	**0.9951**
5°	MPSNR↑	24.9790	24.1003	27.5588	26.2877	26.7749	31.2427	45.0974	41.2673	36.8189	**47.3217**
	RMSE↓	17.6820	21.2049	12.9926	16.4281	13.2530	8.2124	1.6341	2.7275	4.5112	**1.3086**
	ERGAS↓	9.1901	10.0163	7.4100	6.2421	4.5070	2.4096	0.7088	1.0411	1.6957	**0.6527**
	SAM↓	3.9328	14.2359	4.0207	14.0054	4.8585	4.1716	1.9267	2.1527	5.7256	**1.8043**
	UIQI↑	0.4372	0.5645	0.6411	0.7308	0.5831	0.8534	0.9407	0.9338	0.8970	**0.9447**
	MSSIM↑	0.7887	0.6426	0.9006	0.8092	0.6783	0.9644	0.9936	0.9894	0.9647	**0.9945**

TABLE II

EVALUATION OF THE PROPOSED AND COMPARATIVE TECHNIQUES ON THE CHIKUSEI DATASET ENTAILED UTILIZING A GAUSSIAN BLUR KERNEL AND A SCALING FACTOR OF 4. PRESENTED ARE THE MEAN MPSNR, RMSE, ERGAS, SAM, UIQI, AND MSSIM VALUES, WITH ↑ INDICATING SUPERIOR PERFORMANCE WITH LARGER VALUES, AND ↓ SUGGESTING IMPROVED PERFORMANCE WITH SMALLER VALUES. THE TOP-RANKED AND SECOND-BEST OUTCOMES ARE **EMPHASIZED** IN BOLD AND UNDERLINED, RESPECTIVELY

angle	Indices	CNMF [50]	HySure [51]	CSU [4]	u^2MDN [41]	UHIF-RIM [22]	MoG-DCN [18]	MSST [31]	PSRT [32]	MoE-PNP [42]	DUNET
2°	MPSNR↑	30.1478	30.3981	30.9423	38.1211	34.9005	38.5212	40.8396	39.1910	43.5185	**48.6473**
	RMSE↓	9.9493	9.3832	8.5986	16.1821	9.4915	4.4429	4.2887	3.8801	2.4343	**1.5838**
	ERGAS↓	9.4001	10.5306	9.1261	4.9708	3.0957	1.9460	1.6768	1.9591	1.5292	**1.2765**
	SAM↓	6.3829	8.7085	4.7358	24.2999	8.0298	3.5084	3.7163	3.1026	3.0522	**2.0798**
	UIQI↑	0.7119	0.8141	0.7908	0.7232	0.8193	0.9324	0.9571	0.9439	0.9632	**0.9707**
	MSSIM↑	0.8538	0.8540	0.8999	0.7515	0.8161	0.9724	0.9797	0.9750	0.9865	**0.9903**
5°	MPSNR↑	25.6374	22.2822	28.5100	34.2511	34.8777	36.4817	41.1513	33.5536	41.3881	**47.3524**
	RMSE↓	18.5377	26.1713	11.9736	19.2783	10.0247	6.3348	3.7804	6.3638	2.9521	**1.8839**
	ERGAS↓	14.7084	24.8608	11.6276	6.5982	3.1473	2.3806	1.7657	3.9267	1.7687	**1.3834**
	SAM↓	10.3882	18.0102	6.6744	35.0305	8.6094	4.5283	3.7408	5.5896	3.8683	**2.3646**
	UIQI↑	0.4226	0.5587	0.6056	0.6030	0.8101	0.9082	0.9505	0.8541	0.9554	**0.9665**
	MSSIM↑	0.6810	0.6074	0.8283	0.6390	0.8084	0.9646	0.9782	0.9346	0.9813	**0.9892**

TABLE III

EVALUATION OF THE PROPOSED AND COMPARATIVE TECHNIQUES ON THE HOUSTON DATASET ENTAILED UTILIZING A GAUSSIAN BLUR KERNEL AND A SCALING FACTOR OF 4. PRESENTED ARE THE MEAN MPSNR, RMSE, ERGAS, SAM, UIQI, AND MSSIM VALUES, WITH ↑ INDICATING SUPERIOR PERFORMANCE WITH LARGER VALUES, AND ↓ SUGGESTING IMPROVED PERFORMANCE WITH SMALLER VALUES. THE TOP-RANKED AND SECOND-BEST OUTCOMES ARE **EMPHASIZED** IN BOLD AND UNDERLINED, RESPECTIVELY

angle	Indices	CNMF [50]	HySure [51]	CSU [4]	u^2MDN [41]	UHIF-RIM [22]	MoG-DCN [18]	MSST [31]	PSRT [32]	MoE-PNP [42]	DUNET
2°	MPSNR↑	31.9171	35.9280	31.6835	40.0092	33.8053	36.8821	39.1879	36.3038	39.4577	**48.2244**
	RMSE↓	8.3182	6.4057	7.8078	4.1640	7.0649	4.3984	5.0004	4.4305	8.7593	**3.7379**
	ERGAS↓	4.1436	2.8979	4.3749	0.9994	1.7855	1.2301	1.2654	1.7207	2.1075	**0.9652**
	SAM↓	5.5108	4.6342	4.2220	3.4575	5.7846	3.1074	3.1583	**1.7013**	7.9851	3.5536
	UIQI↑	0.8743	0.9201	0.8982	0.9396	0.9067	0.9495	0.9509	0.9654	0.9379	**0.9793**
	MSSIM↑	0.9021	0.9387	0.9392	0.9777	0.8955	0.9805	0.9821	0.9899	0.9798	**0.9931**
5°	MPSNR↑	24.6737	25.5465	26.6294	31.0317	29.2664	39.0586	40.3132	34.1189	40.6214	**47.2561**
	RMSE↓	19.5552	17.1703	14.0772	10.0329	11.5038	4.1007	4.8113	5.7974	6.8594	**3.9029**
	ERGAS↓	9.4382	8.7778	7.8581	2.5841	2.9637	1.1459	1.2464	2.4651	1.5997	**1.0017**
	SAM↓	12.7726	11.4902	7.7186	7.9565	9.2369	3.0521	3.5059	**2.1744**	6.6934	3.6852
	UIQI↑	0.6759	0.7842	0.7443	0.8756	0.8361	0.9548	0.9493	0.9558	0.9444	**0.9713**
	MSSIM↑	0.6839	0.7369	0.8470	0.8878	0.7749	0.9853	0.9858	0.9838	0.9864	**0.9926**

outdoor HSI and remote sensing HSI. This superiority is due to our method's use of supervised end-to-end learning and the integration of hybrid Transformers and spatial-frequency modules to extract spatial and spectral information.

D. Ablation Studies

To simplify the ablation experiments, we rotated the LR-HSI by 2° and then fused the rotated HSI with the MSI.

1) Setting of the Iteration Number N in the Network: The DUNET network consists of N iteration phases. The setting of parameter N is crucial for the performance of DUNET. We assessed the performance of DUNET across various settings of N, as outlined in Table V, and observed that increasing the number of iterations phases leads to a deeper network and improved performance. However, deeper networks require more running time, so we need to strike

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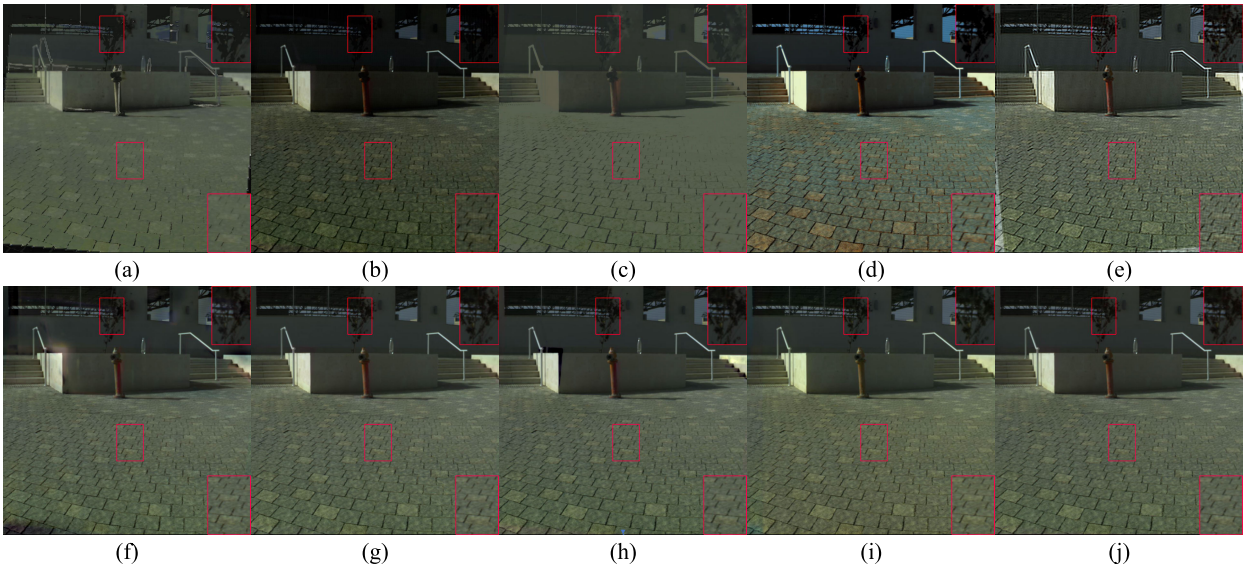


Fig. 3. Fused images of strt_0331-1027 selected from ICVL dataset with pseudocolor composite map (bands 30, 20, and 10). (a) CNMF [50]. (b) HySure [51]. (c) CSU [4]. (d) u^2MDN [41]. (e) UHIF-RIM [22]. (f) MoG-DCN [18]. (g) MSST [31]. (h) PSRT [32]. (i) MoE-PNP [42]. (j) DUNET.

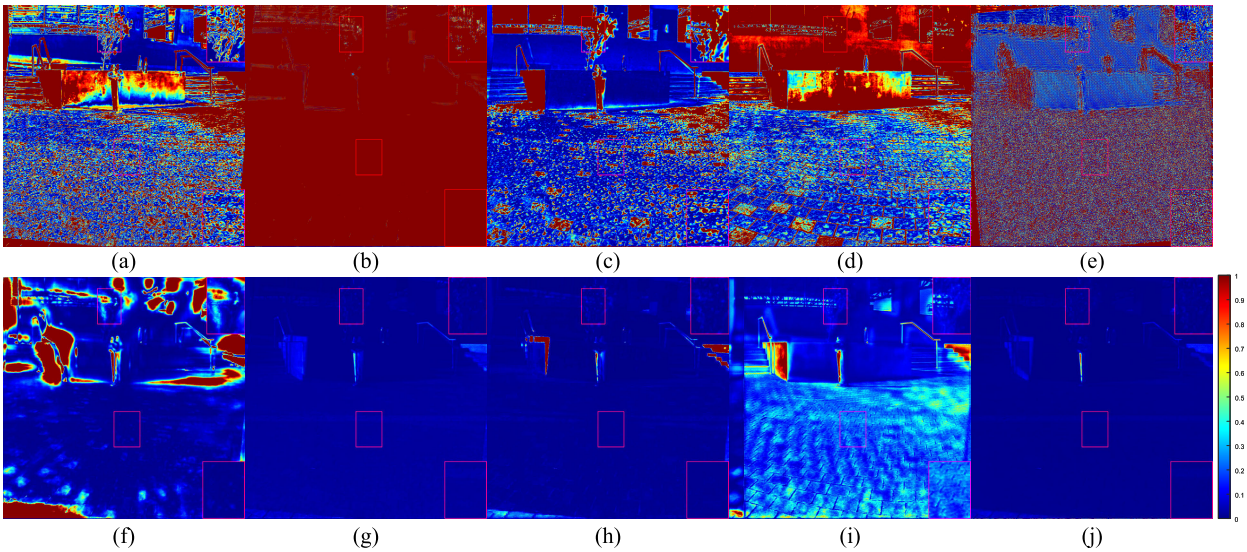


Fig. 4. Comparison methods of fused images of strt_0331-1027 with mean square error images. (a) CNMF [50]. (b) HySure [51]. (c) CSU [4]. (d) u^2MDN [41]. (e) UHIF-RIM [22]. (f) MoG-DCN [18]. (g) MSST [31]. (h) PSRT [32]. (i) MoE-PNP [42]. (j) DUNET.

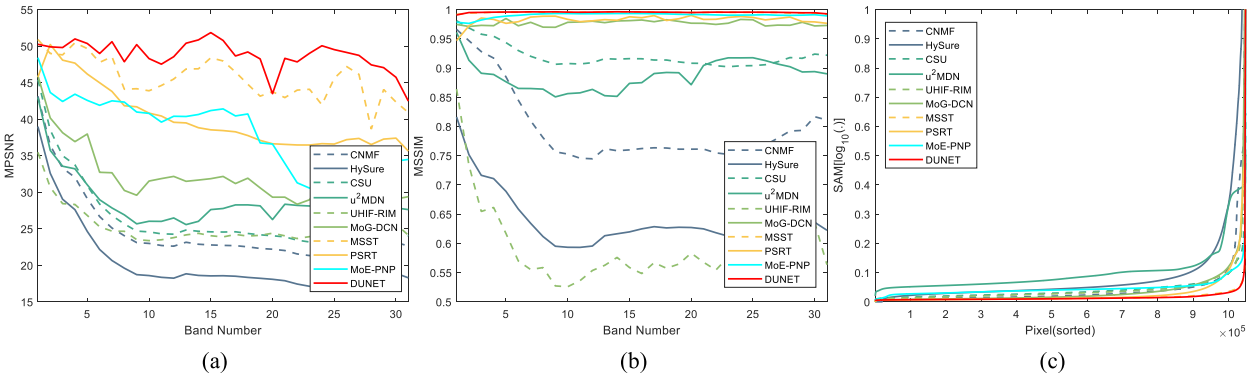


Fig. 5. Corresponding curves of different methods were compared on the strt_0331-1027 data in ICVL, including (a) MPSNR, (b) MSSIM, and (c) SAM.

a balance between performance and runtime. As shown in Table V, the performance is optimal when N is set to 5. Therefore, we set N to 5.

2) Effect of the Weighting Factor μ of S-MSA: We also conducted experiments to explore the effect of the weighting factor μ of S-MSA. μ is used to control the weight of

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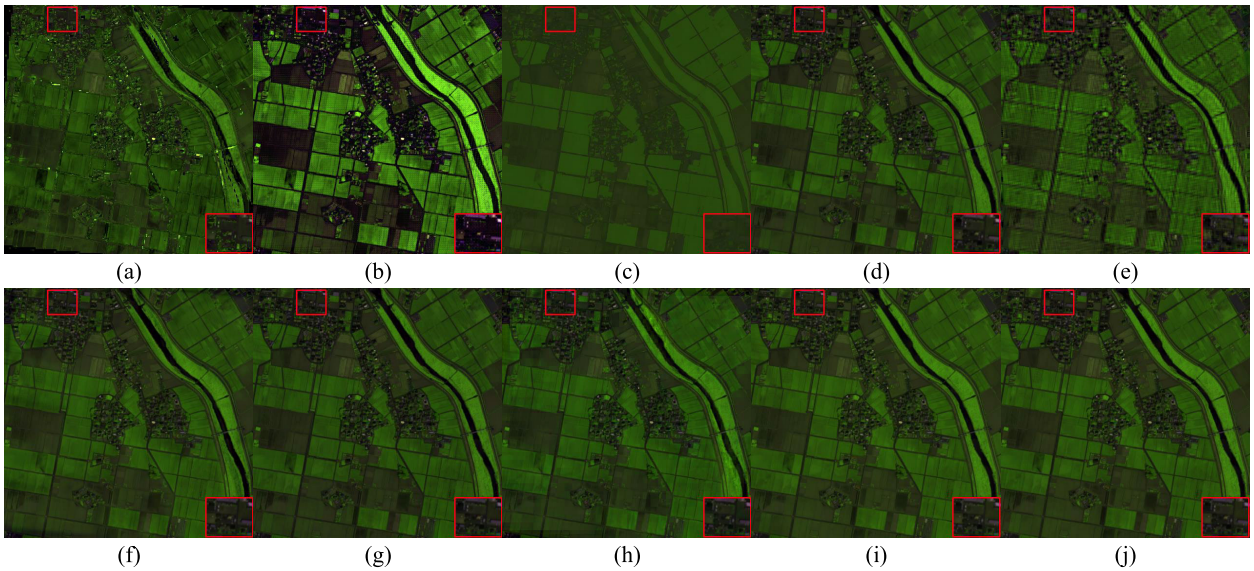


Fig. 6. Fused images of Chikusei with pseudocolor composite map (bands 30, 20, and 10). (a) CNMF [50]. (b) HySure [51]. (c) CSU [4]. (d) u^2MDN [41]. (e) UHIF-RIM [22]. (f) MoG-DCN [18]. (g) MSST [31]. (h) PSRT [32]. (i) MoE-PNP [42]. (j) DUNET.

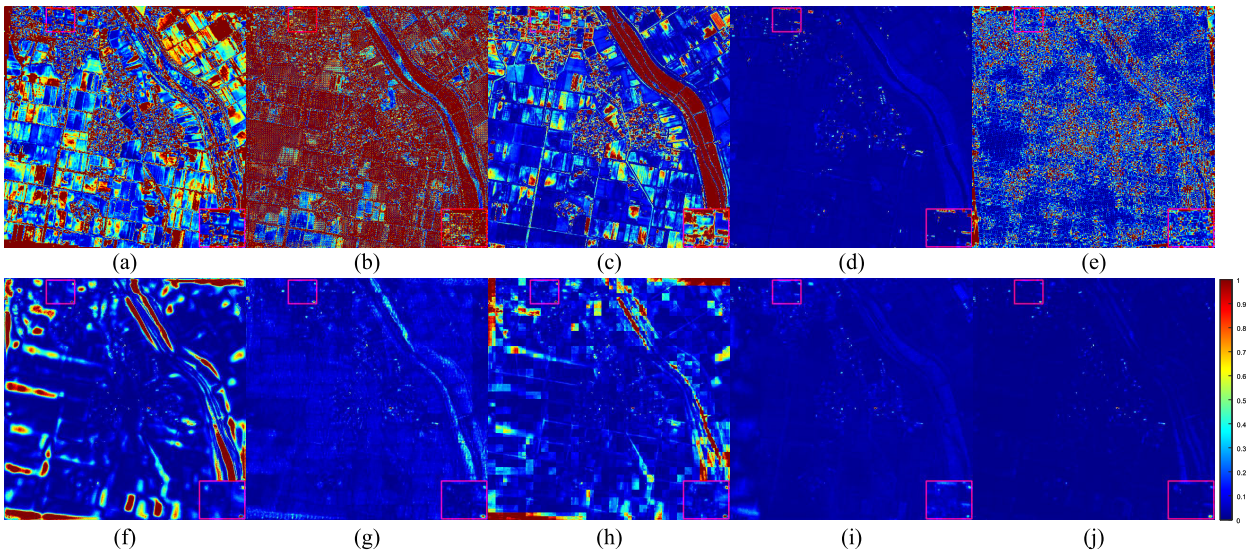


Fig. 7. Comparison methods of fused images of Chikusei with mean square error images. (a) CNMF [50]. (b) HySure [51]. (c) CSU [4]. (d) u^2MDN [41]. (e) UHIF-RIM [22]. (f) MoG-DCN [18]. (g) MSST [31]. (h) PSRT [32]. (i) MoE-PNP [42]. (j) DUNET.

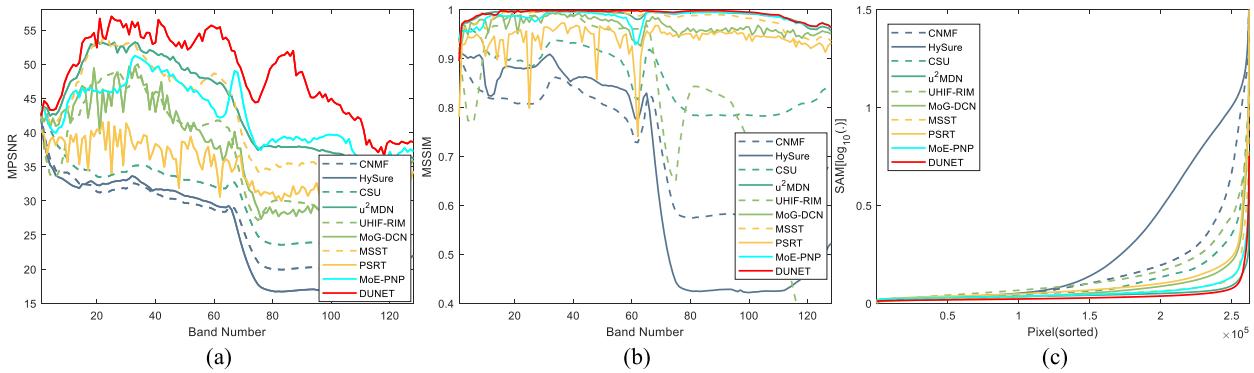


Fig. 8. Corresponding curves of different methods were compared on the Chikusei, including (a) MPSNR, (b) MSSIM, and (c) SAM.

S-MSA features for feature fusion. A larger μ indicates a higher weight of features extracted by S-MSA, while $\mu = 0$ means that S-MSA is not used. Table VI indicates that the model achieved optimal performance with μ set to 0.1. This

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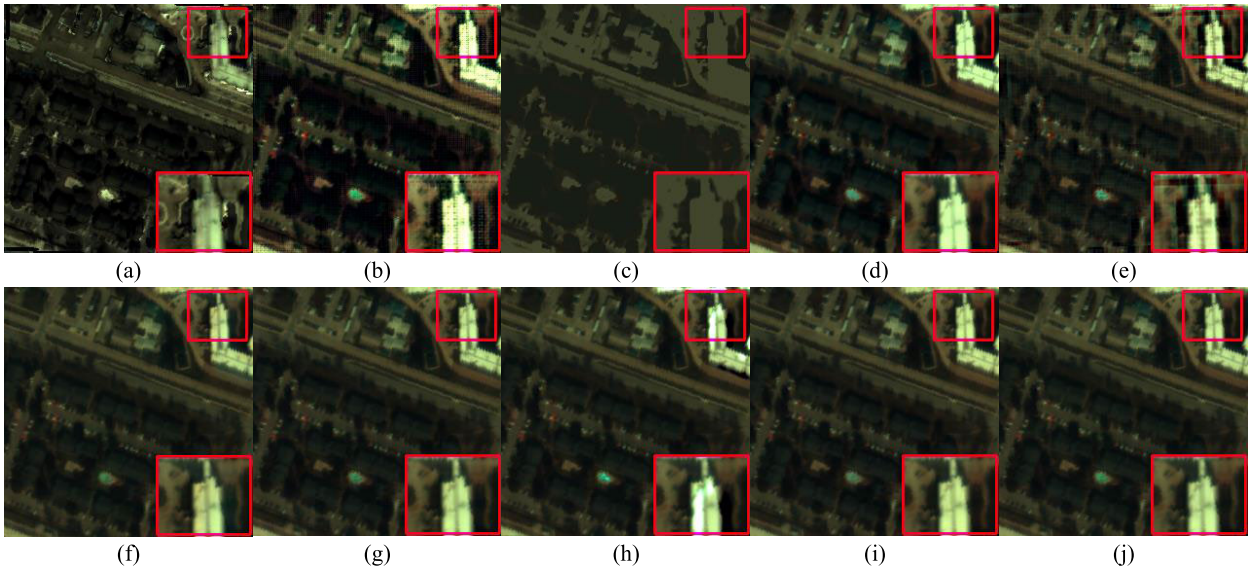


Fig. 9. Fused images of Houston with pseudocolor composite map (bands 20, 10, and 5). (a) CNMF [50]. (b) HySure [51]. (c) CSU [4]. (d) u^2MDN [41]. (e) UHIF-RIM [22]. (f) MoG-DCN [18]. (g) MSST [31]. (h) PSRT [32]. (i) MoE-PNP [42]. (j) DUNET.

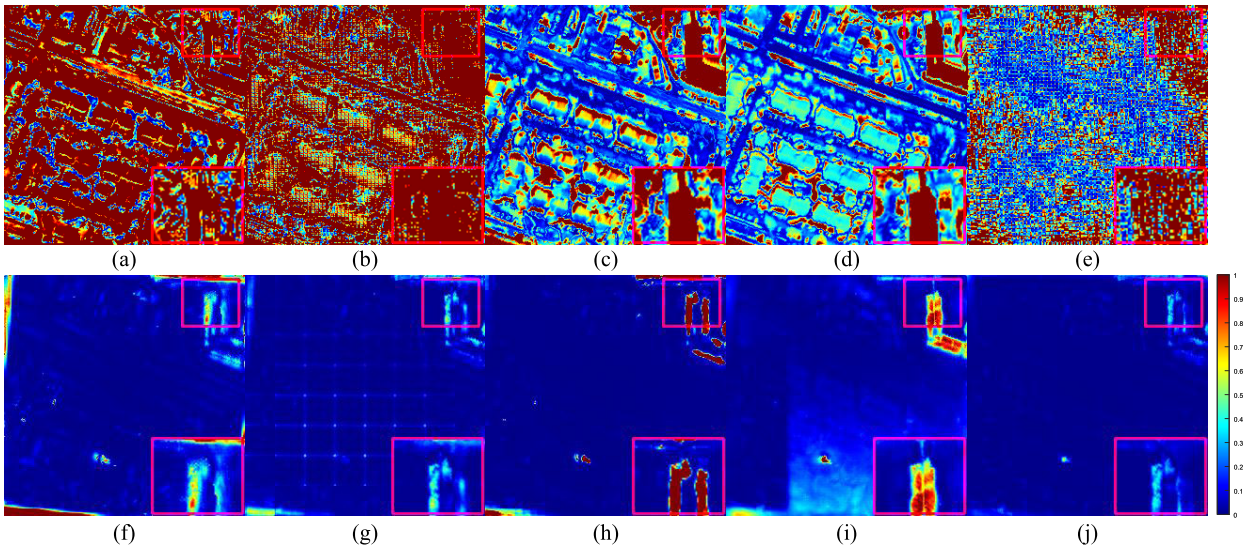


Fig. 10. Comparison methods of fused images of Houston with mean square error images. (a) CNMF [50]. (b) HySure [51]. (c) CSU [4]. (d) u^2MDN [41]. (e) UHIF-RIM [22]. (f) MoG-DCN [18]. (g) MSST [31]. (h) PSRT [32]. (i) MoE-PNP [42]. (j) DUNET.

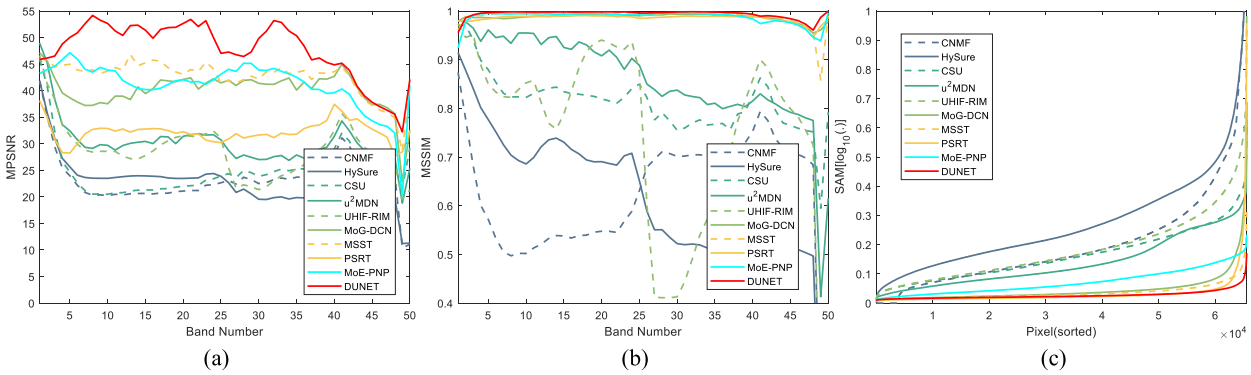


Fig. 11. Corresponding curves of different methods were compared on the Houston, including (a) MPSNR, (b) MSSIM, and (c) SAM.

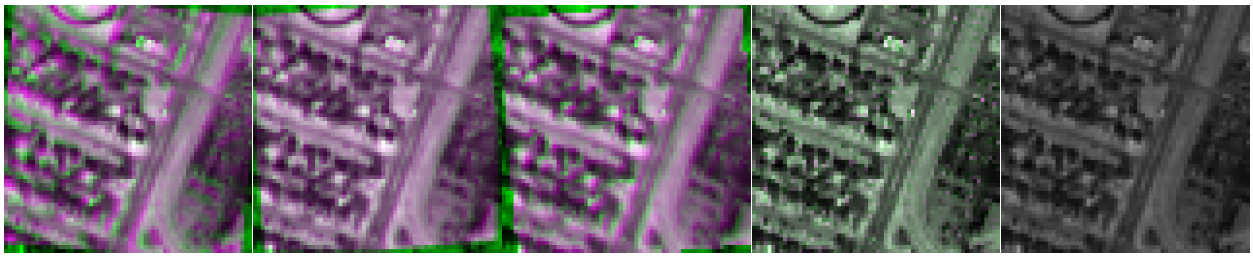
indicates that S-MSA and W-MSA might face optimization issues. Assigning a smaller weighting factor to the S-MSA branch can mitigate this problem, leading to better-combined results.

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(a) (b) (c) (d) (e)

Fig. 12. Visualization of intermediate features, the (20 10 5) band of feature cubes. (a) $\boldsymbol{\mathcal{Z}}_1$. (b) $\boldsymbol{\mathcal{Z}}_2$. (c) $\boldsymbol{\mathcal{Z}}_3$. (d) $\boldsymbol{\mathcal{Z}}_4$. (e) $\boldsymbol{\mathcal{Z}}_5$.



(a) (b) (c) (d) (e)

Fig. 13. Image registration results on the Houston dataset using (a) original image, (b) MI [52], (c) NED [22], (d) our method, and (e) ground truth.

TABLE IV

AVERAGE MPSNR AND MSSIM RESULTS OF THE LR-HSI IMAGES GENERATED BY THE PROPOSED METHOD AND COMPARISON METHODS ON THE ICVL AND HARVARD DATASETS

dataset	angle	Indices	MI [52]	NED [22]	DUNET
ICVL	2°	MPSNR↑	27.9615	28.3790	**36.1015**
		MSSIM↑	0.9263	0.9384	**0.9947**
	5°	MPSNR↑	23.0012	26.2448	**36.6516**
		MSSIM↑	0.7579	0.9016	**0.9944**

TABLE V

THIS STUDY AIMS TO EXPLORE THE IMPACT OF VARYING THE NUMBER OF ITERATIONS ON THE NETWORK PERFORMANCE FOR THE ICVL DATASETS

N	PSNR	RMSE	ERGAS	SAM	UIQI	SSIM
1	46.4211	1.3832	0.7224	1.8484	0.9395	0.9924
3	48.5317	1.1034	0.6041	1.5611	0.9485	0.9949
5	**48.8147**	**1.0786**	**0.5562**	1.5552	**0.9502**	**0.9951**
7	48.6447	1.0999	0.5738	**1.5519**	0.9484	0.9950

TABLE VI

IMPACT OF THE WEIGHTING FACTOR μ IN S-MSA ON THE ICVL DATASET

μ	PSNR	RMSE	ERGAS	SAM	UIQI	SSIM
1	48.4856	1.1076	0.5779	1.6412	0.9478	0.9948
0.1	**48.8147**	**1.0786**	**0.5562**	**1.5552**	**0.9502**	**0.9951**
0.01	48.5208	1.1111	0.5712	1.6277	0.9493	0.9949

3) Selection of Parameter τ_1: Explored in this section is the impact of the parameter τ_1 on the quality indicators of the proposed method. When selecting τ_1 from the set $\{1, 0.1, 0.01\}$ in this experiment, the optimal performance of the proposed method was observed at $\tau_1 = 0.1$, as shown in Table VII. Therefore, we ultimately set τ_1 to 0.1.

TABLE VII

SELECTION OF PARAMETER τ_1 ON THE ICVL DATASET

τ_1	PSNR	RMSE	ERGAS	SAM	UIQI	SSIM
1	48.6437	1.0982	**0.5457**	**1.5319**	0.9499	0.9951
0.1	**48.8147**	**1.0786**	0.5562	1.5552	**0.9502**	**0.9951**
0.01	48.6558	1.1069	0.5621	1.5893	0.9497	0.9950

TABLE VIII

SELECTION OF PARAMETER τ_2 ON THE ICVL DATASET

τ_2	PSNR	RMSE	ERGAS	SAM	UIQI	SSIM
0.5	**48.8147**	**1.0786**	0.5562	**1.5552**	**0.9502**	**0.9951**
0.05	48.6722	1.1016	**0.5426**	1.5999	0.9500	0.9951

4) Selection of Parameter τ_2: In this portion, we explored how adjusting the parameter τ_2 influences the effectiveness of our proposed method, focusing on various quality indicators. For this experiment, we chose values for τ_2 from the set $\{0.5, 0.05\}$. Table VIII presents the results, revealing that the proposed method performs best when τ_2 is set to 0.5. As a result, we decided to adopt τ_2 as 0.5 for our method.

5) Floating-Point Operations and Network Parameter Analysis: Table IX presents the quantitative comparison of the floating point operations per second (FLOPs) and network parameters of different methods on the ICVL dataset. By observing the table, it can be seen that the proposed method achieves excellent results in terms of FLOPs and network parameters compared with the latest registration

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TABLE IX
FLOPS AND NETWORK PARAMETERS OF VARIOUS ALGORITHMS ON THE ICVL DATASET

Indices	CNMF [50]	HySure [51]	CSU [4]	u^2MDN [41]	UHIF-RIM [22]	MoG-DCN [18]	MSST [31]	PSRT [32]	MoE-PNP [42]	DUNET
FLOPs(G)	—	—	—	—	—	12214.13	1999.75	252.01	60816.31	1180.29
Parameters(M)	—	—	—	—	—	7.07	4.37	0.25	44.67	29.85

TABLE X
THIS STUDY AIMS TO EXPLORE THE IMPACT OF VARYING THE NUMBER OF HAT MODULES ON THE NETWORK PERFORMANCE FOR THE ICVL DATASET

L	PSNR	RMSE	ERGAS	SAM	UIQI	SSIM
1	48.6378	1.1052	0.5663	1.5966	0.9487	0.9948
2	48.6019	1.0978	0.5661	1.5785	0.9489	0.9949
3	**48.8147**	**1.0786**	0.5562	**1.5552**	0.9502	**0.9951**
4	48.7294	1.0941	**0.5489**	1.5619	**0.9505**	0.9950

TABLE XI
IMPACT OF FIXED HYPERPARAMETERS AND AUTOMATIC LEARNING ON NETWORK PERFORMANCE

$\{\alpha, \beta, \gamma, \delta, \eta\}$	PSNR	RMSE	ERGAS	SAM	UIQI	SSIM
Automatic Learning	**48.8147**	**1.0786**	0.5562	**1.5552**	**0.9502**	**0.9951**
{0.1,0.001, 0.0001, 0.1, 0.5}	48.5466	1.0902	**0.5386**	1.5910	0.9484	0.9948
{0.01,0.001, 0.0001, 0.1, 0.5}	43.1098	1.9703	0.8392	2.2836	0.9292	0.9896
{0.1,0.01, 0.0001, 0.1, 0.5}	48.5084	1.0943	0.5580	1.5889	0.9483,	0.9948
{0.1,0.001, 0.001, 0.1, 0.5}	47.9154	1.2028	0.5651	1.6435	0.9478	0.9946
{0.1,0.001, 0.0001, 0.01, 0.5}	47.8616	1.1706	0.6253	1.6515	0.9449	0.9942
{0.1,0.001, 0.0001, 0.1, 0.05}	36.1166	4.7176	1.4170	3.1147	0.8927	0.9763

fusion methods. However, there is still room for improvement compared with the PSRT method.

6) Setting of the Number of HAT Modules L in the Network: The prior in the DUNET network consists of L HAT modules. The setting of parameter L is crucial for the performance of DUNET. We assessed the performance of DUNET with different settings of L, as shown in Table X, and observed that increasing the number of HAT modules leads to a deeper network and improved performance. As shown in Table X, the performance is optimal when L is set to 3.

7) Impact of Fixed Hyperparameters and Automatic Learning on Network Performance: Table XI illustrates the impact of fixed hyperparameters and automatically learned hyperparameters on network performance. The fixed hyperparameters are set as α, β, γ, δ, and η as shown in the table. The results show that automatically learned hyperparameters outperform the fixed ones. Determining fixed hyperparameters often requires extensive experimentation and fine-tuning to find suitable values, and different tasks or datasets may require different parameter configurations. In contrast, automatically learned hyperparameters can adjust dynamically based on feedback during training, allowing the model to achieve optimal performance across various scenarios.

V. CONCLUSION

In satellite remote sensing, HSI and MSI possess complementary characteristics, necessitating their fusion to enhance both spatial and spectral resolutions. However, due to different acquisition conditions, there is an inherent misalignment between these datasets, posing a significant challenge. This article proposes a novel approach called the DUNET to address the simultaneous registration and fusion of HSI and MSI without the need for preregistration. Unlike conventional deep fusion methods, DUNET employs deep unfolding, leveraging mutual information and deep priors to facilitate a more optimal degradation model-informed fusion process. The network integrates HATs and spatial-frequency modules to fully exploit the spatial–spectral information of HSI, resulting in a more accurate and detailed scene representation. Through extensive quantitative and visual experiments on three benchmark HSI datasets, we demonstrate the superior performance of DUNET compared with the existing mainstream algorithms, highlighting its effectiveness in remote sensing image fusion.

Given that Mamba [53] is gradually gaining application in the field of computer vision and its performance is comparable to a Transformer but with lower spatial computational complexity, we plan to integrate Mamba into HSI fusion.

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